

Forward pass logic

$$y^* = \text{softmax}(z)$$

$$L = - \sum_i y_i \log y_i^*$$

$$\delta_{\text{logits}} = \frac{\partial L}{\partial z} = y^* - y$$

Expanding the loss

$$L = - \sum_i y_i \log y_i^*$$

Now take partial derivative w.r.t. z_k :

$$\frac{\partial}{\partial z_k} (y_i \log y_i^*) = y_i \frac{\partial}{\partial z_k} (\log y_i^*) \quad (i)$$

Recalling the chain rule of derivatives:

$$\frac{d}{dx} (\log u) = \frac{d \log u}{du} \cdot \frac{du}{dx}$$

Plugging this back into equation i:

$$\begin{aligned} \frac{\partial (y_i \log y_i^*)}{\partial z_k} &= y_i \cdot \frac{1}{y_i^*} \cdot \frac{\partial y_i^*}{\partial z_k} \\ \frac{\partial (y_i \log y_i^*)}{\partial z_k} &= y_i \cdot \frac{1}{y_i^*} \cdot y_i^* (\delta_{ik} - y_k^*) \\ \frac{\partial (y_i \log y_i^*)}{\partial z_k} &= y_i (\delta_{ik} - y_k^*) \\ \frac{\partial (y_i \log y_i^*)}{\partial z_k} &= y_i \delta_{ik} - y_i y_k^* \end{aligned}$$

Now the function

$$\frac{\partial L}{\partial z_k} = - \sum_i (y_i \delta_{ik} - y_i y_k^*)$$

Now let's split the sum:

$$- \sum_i y_i \delta_{ik} + \sum_i y_i y_k^*$$

Since $\delta_{ik} = 1$ if and only if $i = k$ (and 0 otherwise), the first summation collapses to a single term where the index i is replaced by k :

$$\sum_i y_i \delta_{ik} = y_k$$

y is a one-hot vector so all its values sum to 1

Now let's take a look at the second term

$$\begin{aligned} \sum_i y_i y_k^* \\ y_k^* \sum_i y_i \end{aligned}$$

Now because of one hot encode giving us y_k :-

$$\sum_i y_i = 1$$

$$-y_k + y_k^* = y_k^* - y_k = \frac{\partial L}{\partial z_k}$$

derived in [flashback.md](#)

$$J(y^*) = \text{diag}(y^*) - y^* y^{*T}$$

$$\frac{\partial y_i^*}{\partial z_k} = J_{ik}$$

Now lets take a look at lm head

This is a linear transformation task.

Many years ago I might have used this in the output layer of ANN when doing backpropagation from scratch in java.

Like in the derivation of FlashAttention, we will look at the matrix version of the chain rule to get a better idea of why the transposes appear.

$$Y = AX$$

$$\frac{\partial L}{\partial X} = A^T \frac{\partial L}{\partial Y}$$

$$\frac{\partial L}{\partial A} = \frac{\partial L}{\partial Y} X^T$$

$$z = Wh + b$$

Now gradient with respect to W:

$$\frac{\partial z_i}{\partial W_i} = h$$

$$\frac{\partial L}{\partial W_i} = \frac{\partial L}{\partial z_i} \cdot \frac{\partial z_i}{\partial W_i}$$

From the above equations:

$$\frac{\partial L}{\partial W_i} = (y^* - y)h$$

Stacking all the rows:

$$\frac{\partial L}{\partial W} = (y^* - y)h^T$$

Now, the gradient with respect to b:

$$\frac{\partial z_i}{\partial b_i} = 0 + \frac{\partial b_i}{\partial b_i}$$

$$\frac{\partial z_i}{\partial b_i} = 1$$

$$\frac{\partial L}{\partial b} = y^* - y$$

Similarly, the gradient with respect to h:

$$z = Wh + b$$

$$\frac{\partial z}{\partial h} = W$$

$$\frac{\partial L}{\partial h} = \frac{\partial L}{\partial z} \frac{\partial z}{\partial h}$$

$$\frac{\partial L}{\partial h} = W^T \frac{\partial L}{\partial z}$$

Finally,

$$\delta = y^* - y$$

$$\frac{\partial L}{\partial z} = \delta$$

For weights:

$$\frac{\partial L}{\partial W} = \delta h^T$$

For bias:

$$\frac{\partial L}{\partial b} = \delta$$

For h :

$$\frac{\partial L}{\partial h} = W^T \delta$$

Now `output_proj` inside of the attention head is also a linear transformation.

$$z = Wx + b$$

Upstream gradient:

$$\delta = \frac{\partial L}{\partial z}$$

$$\frac{\partial L}{\partial W} = \delta x^T$$

$$\frac{\partial L}{\partial x} = W^T \delta$$

$$\frac{\partial L}{\partial b} = \delta$$

The linear layer does not generate the upstream gradient; rather, it gets passed along. For example, for $O = PV$, the gradient originates from the Cross-Entropy loss and the softmax function.